School of Electrical Engineering & Telecommunications

ELEC1111 Workshop 6

Q1 (2020 T1 Final Exam Q3a)

In the circuit of Figure 1, calculate the voltage v_0 if $v_s = 2$ V.

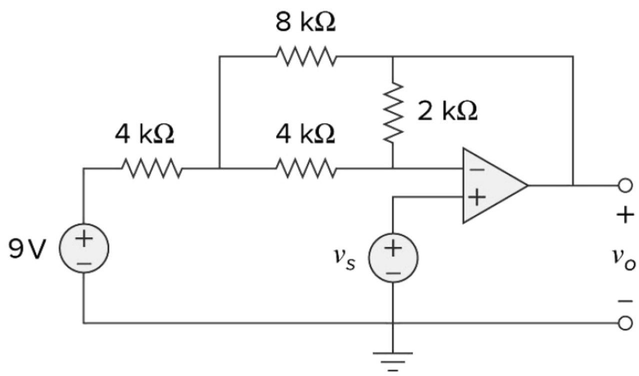

Figure 1

Answer: v_0 = 0.727 V

Q2 (2021 T1 Final Exam Q3a)

Resistance temperature detectors (RTDs) are sensors used to measure temperature based on the resistance change. The most common type (PT100) has a resistance of 100 Ω at 0ºC and 138.4 Ω at 100ºC. The circuit shown in Figure 2 is used to convert the information from the RTD into something meaningful for an analog-to-digital converter (ADC).

i) Calculate the voltage v_0 as a function of v_{in} when the RTD has a value of 130 Ω.

ii) How large can be the value of v_{in} before the amplifier saturates?

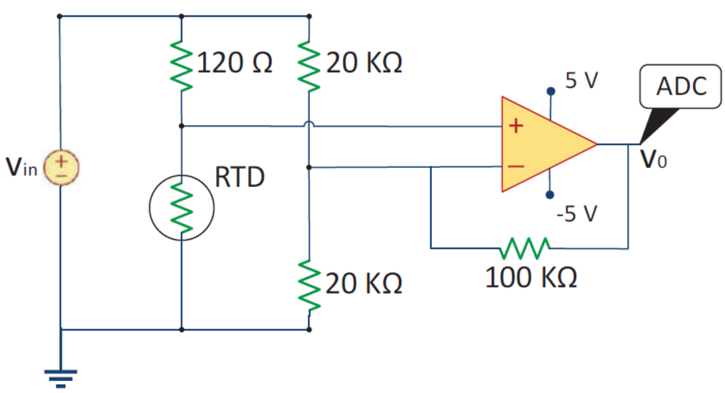

Figure 2

Answers: i) $v_0 = 0.72 v_{in}$, ii) $|v_{in}| = 6.94\ V$

Q3 (2023 T1 Final Exam Q3a)

For the circuit shown in Figure 3,

i) Find i_a.

ii) Find the value of the right voltage source (i.e., v_2) for which $i_a = 0$.

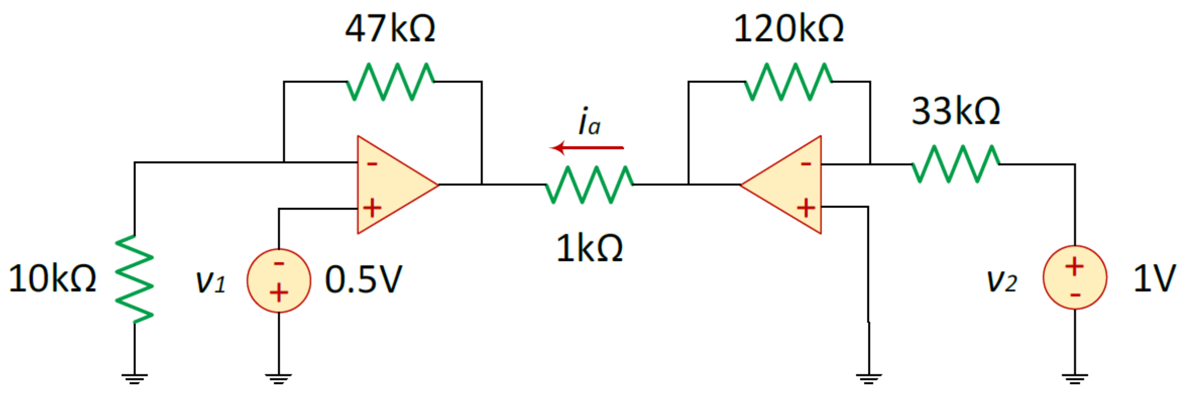

Figure 3

Answers: i) $i_a = -0.79$ mA, ii) $v_2 = 0.783$ V.

Q4 (2022 T1 Final Exam Q3a)

Calculate i_0 in the op amp circuit shown in Figure 4.

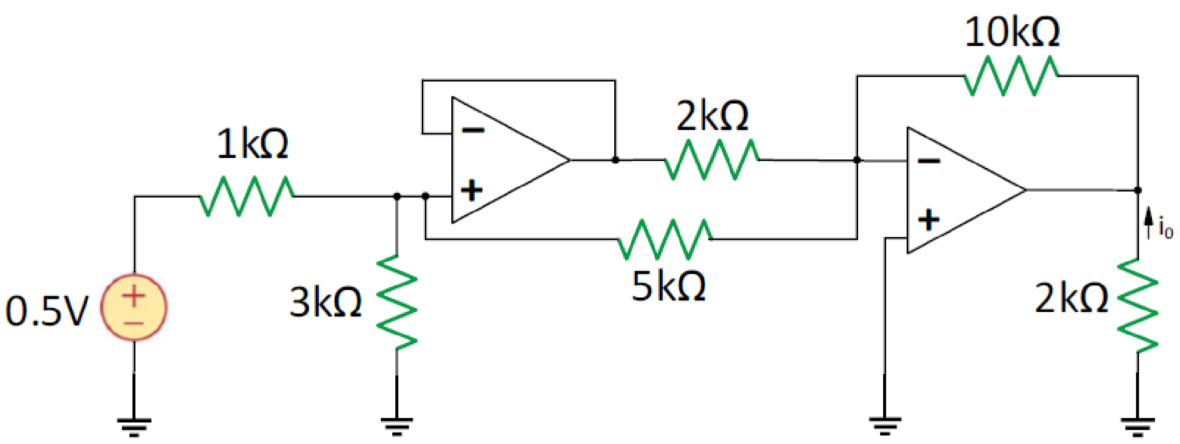

Figure 4

Answer: $i_0 = 1.141\ mA$